

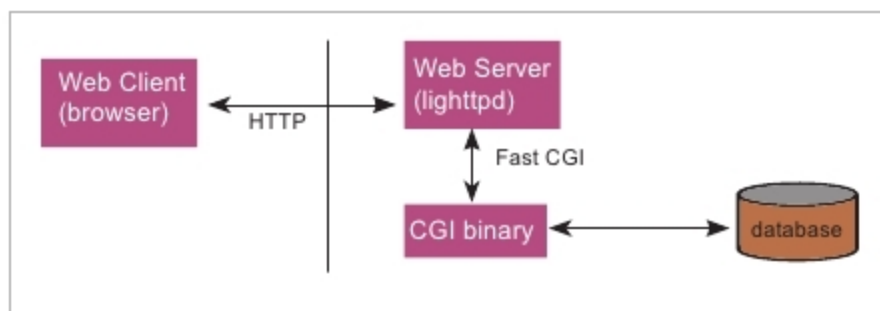


Figure 1: Web server with CGI architecture

```

$ sudo apt-get install autogen autoconf libtool /*these are
supporting programs in host PC*/
$ cd fcgi2-master
$ ./autogen.sh
$ ./configure

```

Note: As before, CC and LD variables have to be set and the necessary options can be given for embedded cross targets.

```

$ make
$ sudo make install

```

Note: As before, collect the binaries and place them in a file system for cross targets.

Step 3: Now we are ready to write and execute our first CGI program. But before starting, edit *lighttpd.conf* for linking the CGI binary.

```

fastcgi.debug = 1
fastcgi.server = (
    "/hellotest" => (
        #"host" => "127.0.0.1",
        #"port" => 51000,
        "bin-path" => "/home/sganguly/Documents/hello_cgittest.
fcgi",
        "socket" => "/tmp/hello_cgittest.sock",
        "check-local" => "disable",
        "max-procs" => 2,
    ))
)

```

Here *hello_cgittest.fcgi* is the name of the binary executable. Socket will be used between the HTTP server and the CGI binary. To get this page, type *127.0.0.1:80/hellotest* in a browser.

```

#include <fcgi_stdio.h>
#include <stdlib.h>
#include <string.h>

```

```

int main (void) {
    unsigned char val = 0;
    char *env;

    while(FCGI_Accept() >= 0){
        printf("Status: 200 OK\r\n");
        printf("Content-type: text/html\r\n\r\n");
        printf("<!doctype><html>");
        printf("<head><meta http-equiv=\"refresh\"
content=\"3\"</head> <body>");
        printf("val %d<br>", val);

        env=getenv("HTTP_USER_AGENT");
        if(env != NULL) {
            printf("User Agent = %s<br>", env);
        }

        env=getenv("CONTENT_TYPE");
        if(env != NULL) {
            printf("Content Type = %s<br>", env);
        }

        printf("</body></html>\n");
        val++;
    }

    return EXIT_SUCCESS;
}

```

FCGI_Accept() will be true with any GET or POST request. Lighttpd also sets some environment variables to be accessible from the CGI code.

Step 4: Let's compile this code:

```

sganguly@breakout:~/Documents$ gcc hello_cgittest.c -o hello_
cgittest.fcgi -lfcgi -Wl,-rpath /usr/local/lib
/usr/local/lib is the location where fcgi library is located
after installation.

```

Once you open the page from the browser, it will look like what's shown in Figure 2.

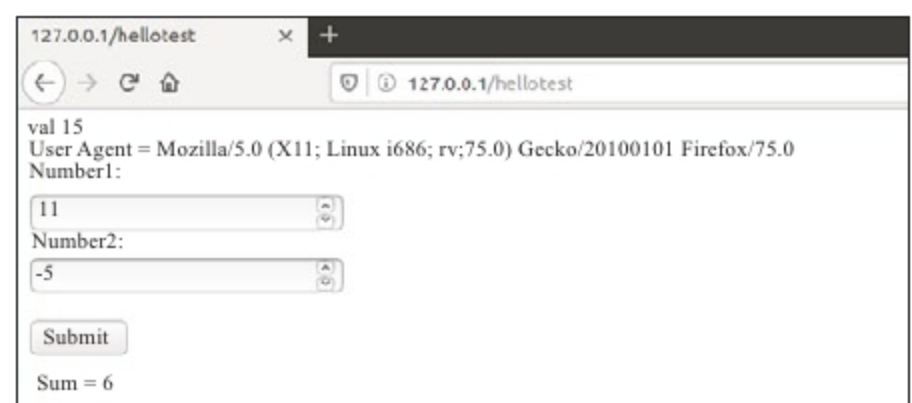


Figure 2: A sample Web page with form (as per code)